

Santiago Silva Avalos

Data Scientist

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in [santiago-silva-avalos](#)

Data Scientist (MSc Math) experienced in building and deploying predictive models, handling end-to-end data engineering tasks, and delivering actionable product insights. I regularly engage stakeholders and clearly communicate complex findings, ensuring alignment with business goals. Comfortable collaborating closely with cross-functional teams, I enjoy fast-paced environments where quick iterations drive progress. I'm skilled at understanding business and client needs, and particularly strong at simplifying technical ideas to keep everyone on the same page.

Experience

- December 2024 – Present **Data Consultant, DD360 – Real Estate Fintech**, Mexico, Countrywide
- Engineered **robust data homology algorithms** to standardize the **cadastral database** and developed a **matching methodology** aligning it with **online real estate listings** to validate and update property records.
 - Leveraged **online property prices** to assess and update **unitary land values** across multiple municipalities, ensuring **fair taxation** and enabling **optimal policymaking**.
 - Redesigned the **ER diagram** for the company's **cadastral services**, establishing a robust, **cartographic source-agnostic architecture**.
 - Delivered **presentations** to communicate the **solution scope and benefits** to **municipal stakeholders**.
- January 2024 – December 2024 **Data Scientist, DD360 – Real Estate Fintech**, Mexico City, MX
- Developed and scaled DD360's v10 **real estate pricing model**, which accounted for **80% of the tech division's revenue**.
 - Expanded coverage to 20 new cities, applying **spatial analysis** (Uber's H3 spatial index) and heterogeneous effects, reducing predictive error by 50%.
 - Led performance marketing optimization using **causal inference** and ML-driven experimentation, achieving **20X lead increase** for participating users.
 - Designed **AI-powered virtual assistant** using **RAG**, improving lead follow-up efficiency for a key real estate firm.
 - **Managed junior data scientist** in developing a **computer vision model** for 900,000+ properties using advanced embeddings.

Education

- October 2022 – August 2023 **MSc Mathematics (Distinction)**, *University of Essex*, Colchester, UK
- Specialized in Machine Learning & Bayesian Statistics.
 - Collaborated on MCMC simulation for Mixed Marketing Modeling.
- August 2019 – August 2023 **BSc Applied Mathematics (GPA: 8.76)**, *Instituto Tecnológico Autónomo de México (ITAM)*, Mexico City, MX
- Focused on Optimization Methods & Statistical Programming
 - Developed Mixed Marketing Model using R and Bayesian inference.

Technical Skills

Core	Python, SQL, TensorFlow, PyTorch, Google Cloud, AWS
Data Science	Causal Inference, Bayesian Statistics, Machine Learning, Geospatial Analysis
Data Engineering	ETL Pipelines, Serverless, Optimization